Hardware Security
硬體安全

Statistical Characteristics of Power Traces

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Outline

- Composition of Power Traces
- Characteristics of Single Point
- Leakage of Single Points
- Characteristics of Power Traces
- Compression of Power Traces
- Confidence Intervals and Hypothesis Testing

Composition of Power Traces

- Each point of a trace can be modeled as

$$P_{total} = P_{op} + P_{data} + P_{el.noise} + P_{const}$$

  - Operation-dependent component
  - Data-dependent component
  - Electronic noise
  - Constant power

Characteristic of Single Points

- 10000 power traces of moving the data value 0 from internal memory to register

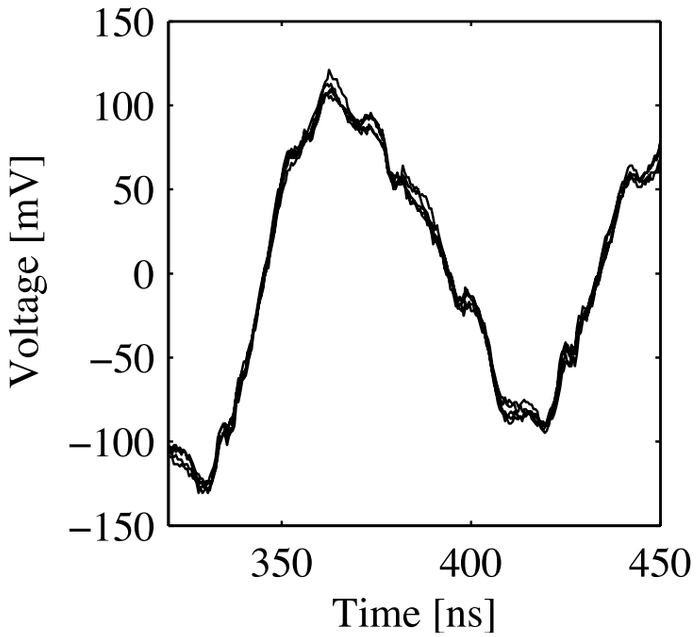


Figure 4.1. Power traces look very similar if the same data is processed.

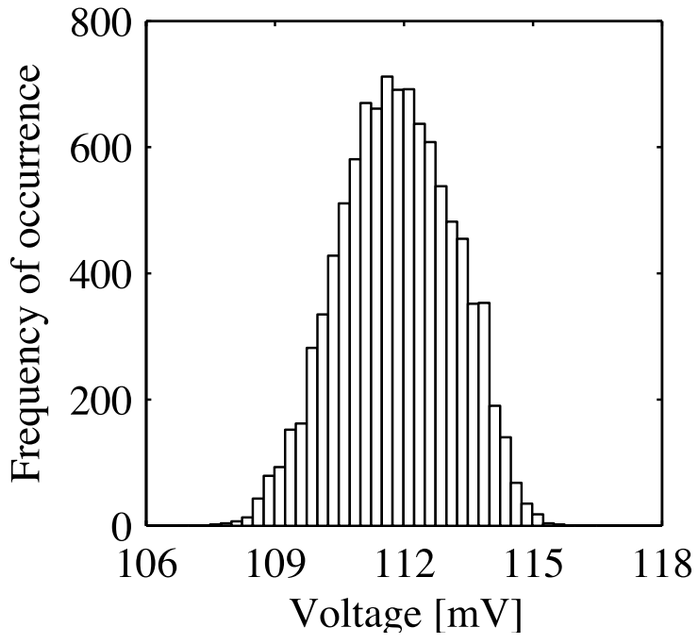


Figure 4.2. Histogram of the power consumption at 362 ns of Figure 4.1.

- Normally distributed electronic noise

$$P_{el.noise} \sim \mathcal{N}(0, \sigma)$$

Normal Distribution

- **PDF** $f(x) = \frac{1}{\sqrt{2\pi}\sigma} \cdot e^{-\frac{1}{2}\cdot\left(\frac{x-\mu}{\sigma}\right)^2}$
- **Estimators**
  - mean $\bar{x} = \frac{1}{n}\sum_{i=1}^{n} x_i$, variance $s^2 = \frac{1}{n-1}\sum_{i=1}^{n}(x_i - \bar{x})^2$

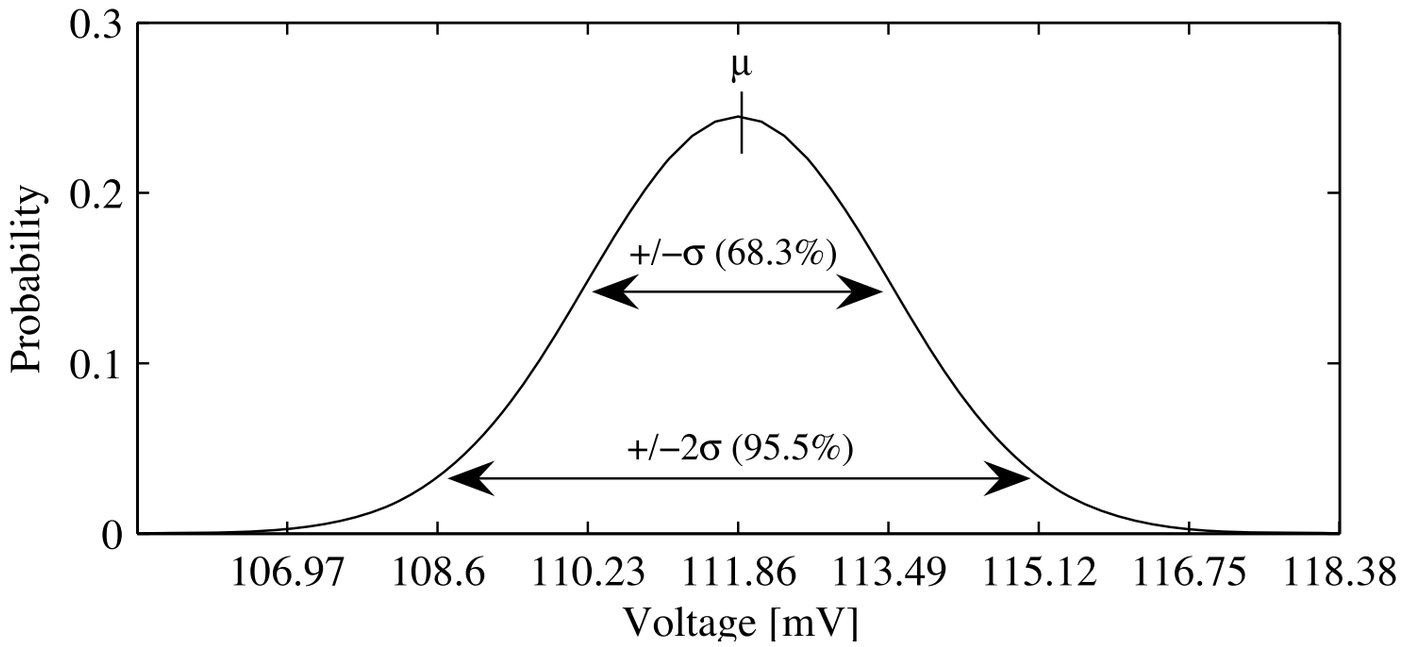


Figure 4.3. The normal distribution $\mathcal{N}(111.86, 1.63)$ models the power consumption at 362 ns

Data Dependency

- 51200 power traces of moving 256 different values from memory to register

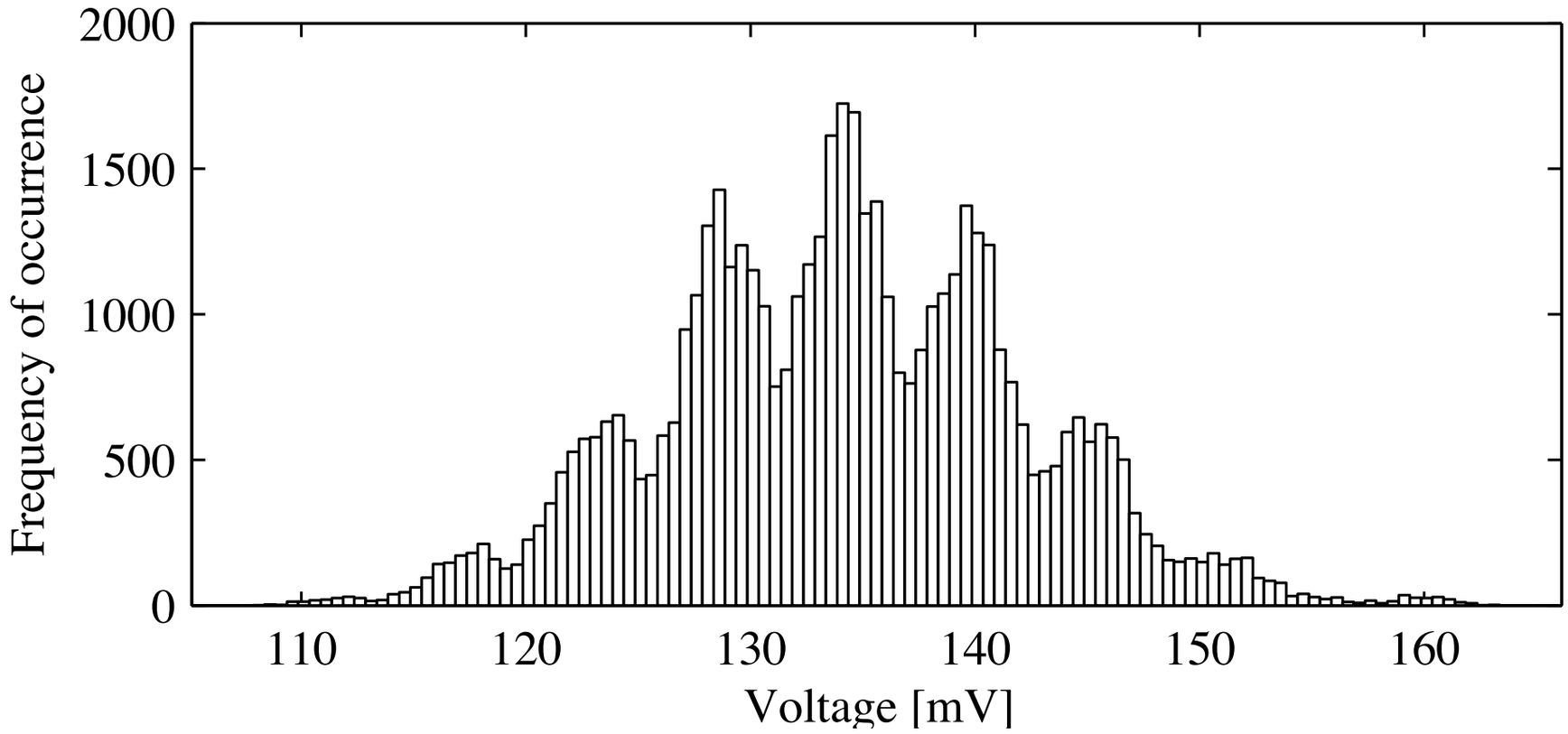


Figure 4.4. Histogram of the power consumption at 362 ns if different data values are transferred from the internal memory to a register.

Table 4.1. Probability distribution for the Hamming weight of a uniformly distributed 8-bit value.

HW	0	1	2	3	4	5	6	7	8
Prob.	0.004	0.031	0.109	0.219	0.273	0.219	0.109	0.031	0.004

Data Dependency

- Inversely proportional to Hamming weight

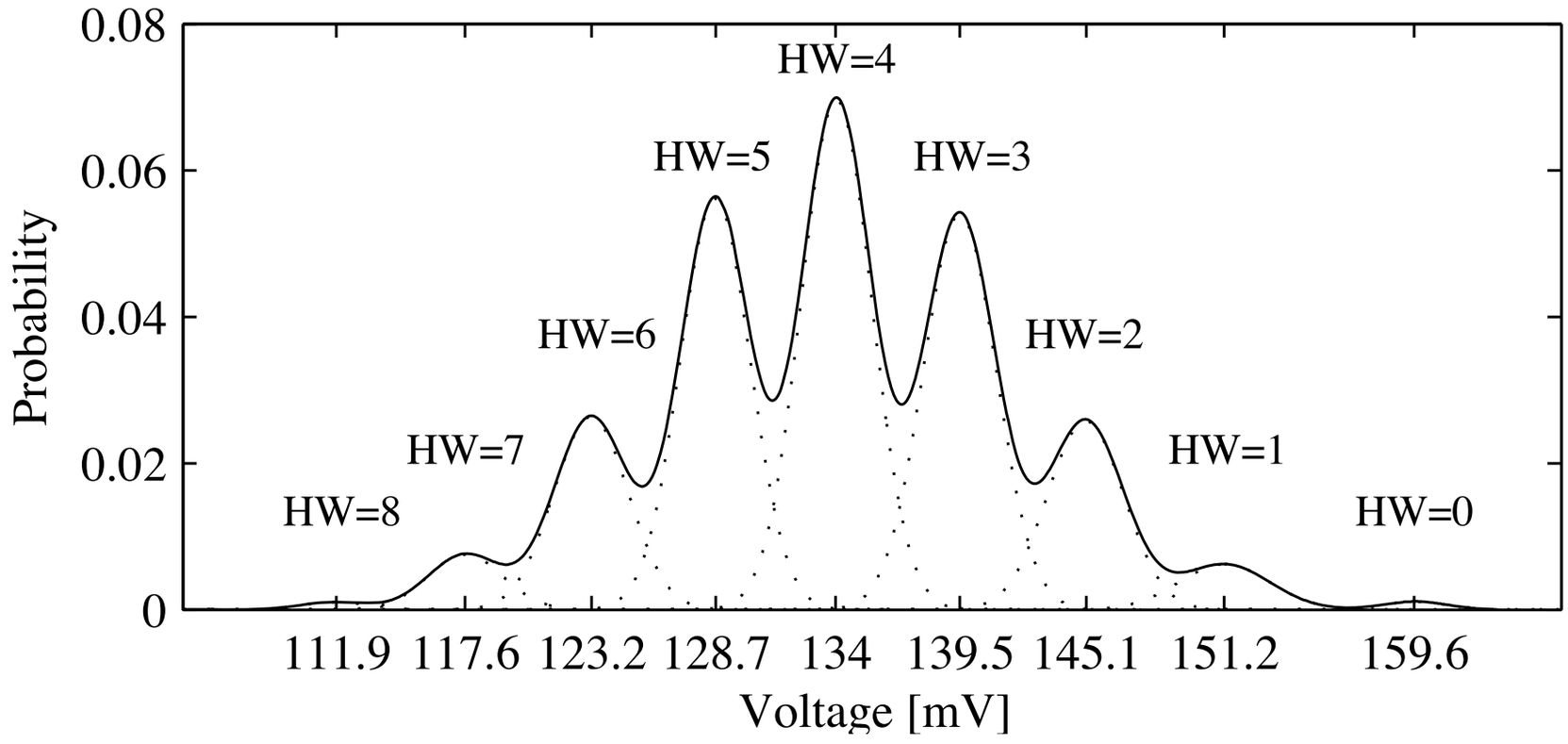


Figure 4.5. The distribution of the power consumption when the microcontroller transfers different data from the internal memory to a register.

$$E(P_{data}) = 0$$

HW	8	7	6	5	4	3	2	1	0
mV	-22.67	-16.92	-11.35	-5.86	-0.49	4.96	10.53	16.68	25.12

Operation Dependency

- Operation dependent power consumption can usually be approximated by a normal distribution.
- Operation and data dependencies may or **may not be independent.**
  - In MCU, they are usually independent; in hardware implementations, they are more likely dependent.

Leakage of Single Points

- **Signal and noise**
  - Exploitable v.s. non-exploitable parts

$$P_{op} + P_{data} = P_{exp} + P_{sw.noise}$$

$$P_{total} = P_{exp} + \underbrace{P_{sw.noise} + P_{el.noise}}_{\text{total noise}} + P_{const}$$

signal

$$SNR = \frac{Var(P_{exp})}{Var(P_{sw.noise} + P_{el.noise})}$$

How to Compute SNR from Traces

$$\mathrm{SNR} = \frac{\mathrm{Var}[\mathrm{E}[L|X]]}{\mathrm{E}[\mathrm{Var}[L|X]]}$$

1. Group traces according to the targeted value
2. Compute the average and variance traces of each group
3. Compute the variance of the average traces of each group (which is the *signal*)
4. Compute the average of the variance traces of each group (which is the *noise*)

SNR of Single Points

- At 362 ns

Component	Variance 8-bit scenario	Variance 1-bit scenario
P_{data}	61.12	61.12
P_{op}	0.00	0.00
$P_{el.noise}$	2.67	2.67
P_{exp}	61.12	6.87
$P_{sw.noise} + P_{el.noise}$	2.67	56.85

Avg. power	consumption
LSB=0	131.49mV
LSB=1	136.73mV

- SNR: 22.89 v.s. 0.12

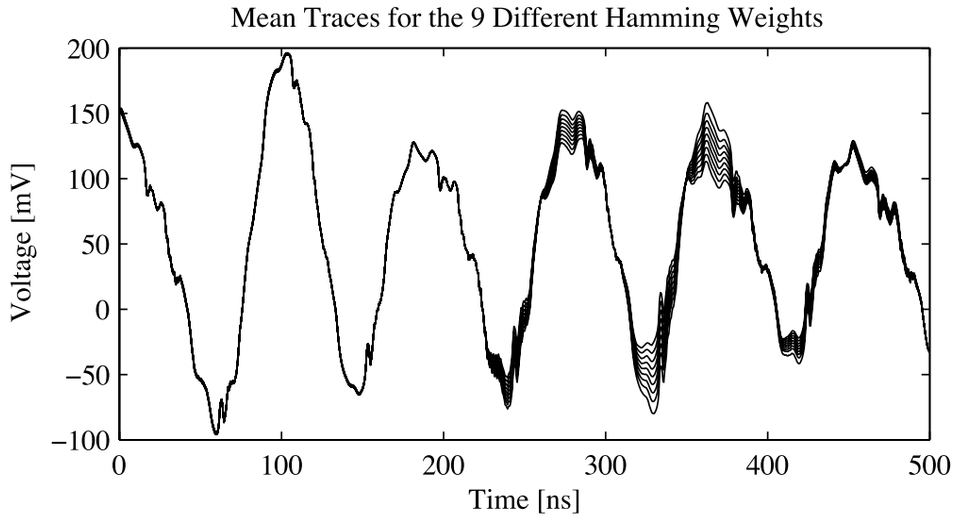
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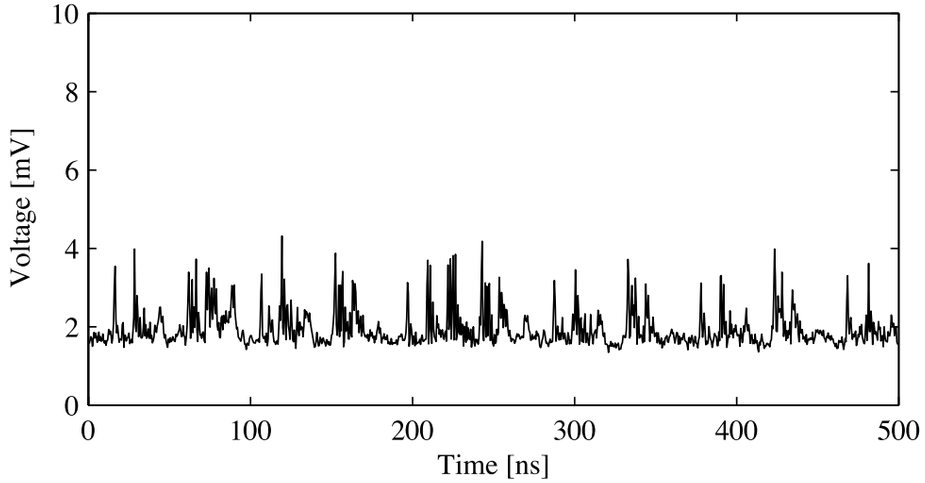
HW	0	1	2	3	4	5	6	7	8
Prob.	0.004	0.031	0.109	0.219	0.273	0.219	0.109	0.031	0.004

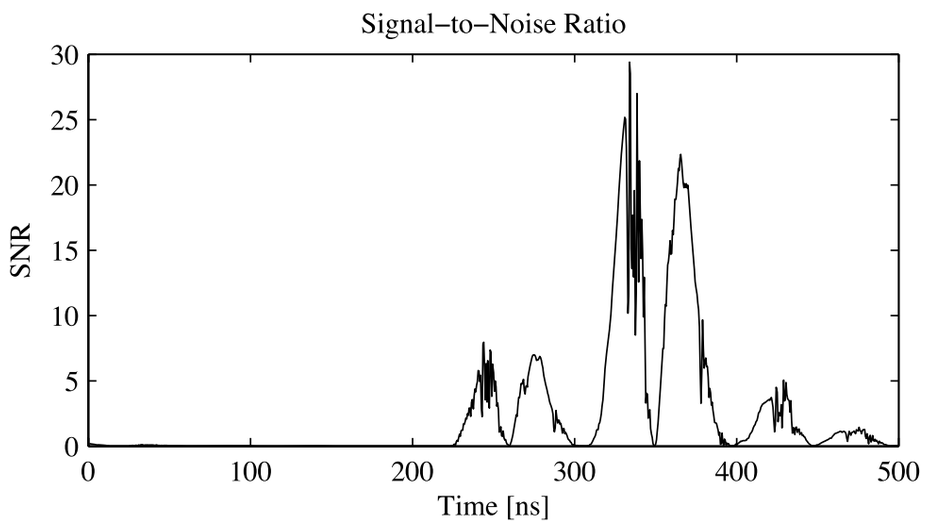
HW	8	7	6	5	4	3	2	1	0
mV	-22.67	-16.92	-11.35	-5.86	-0.49	4.96	10.53	16.68	25.12

SNR of Neighboring Points

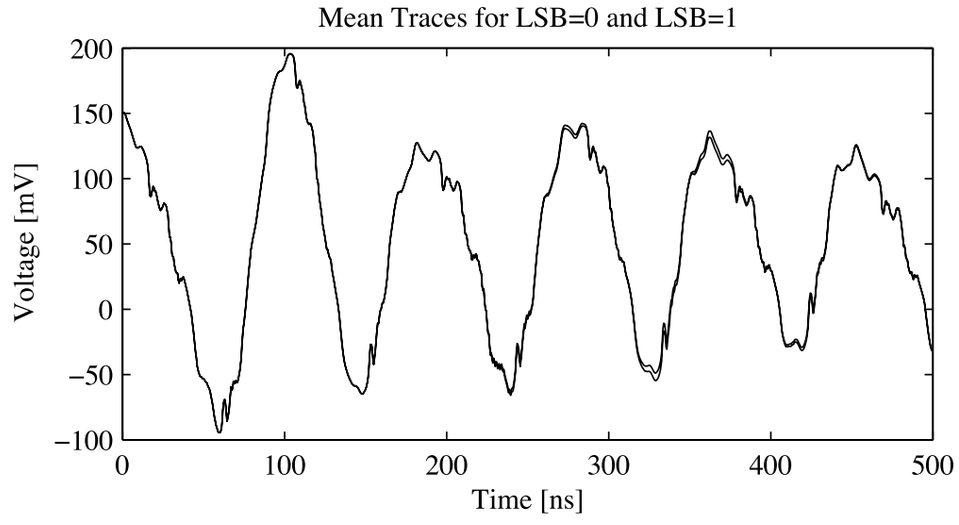
8-bit

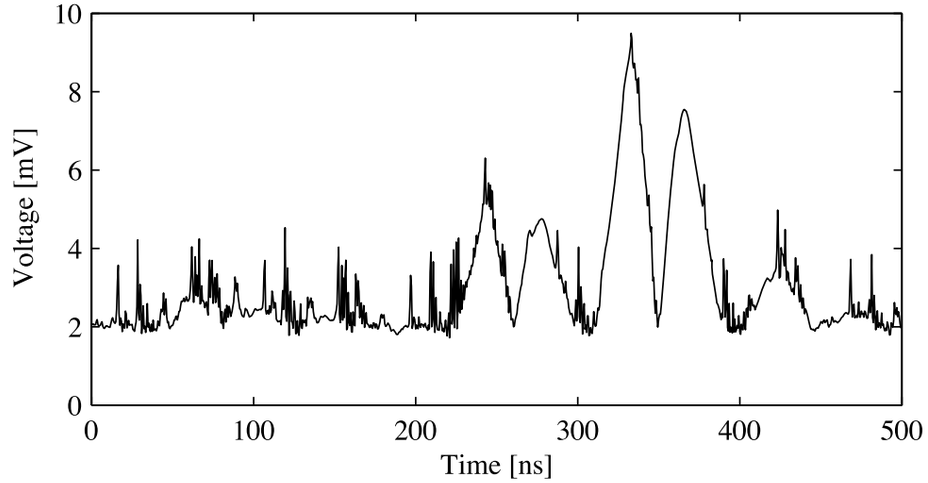


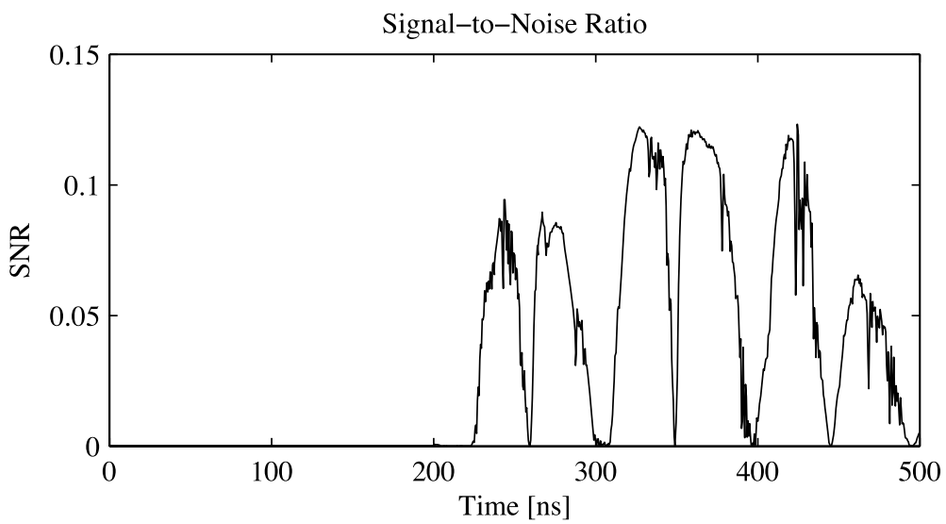




1-bit

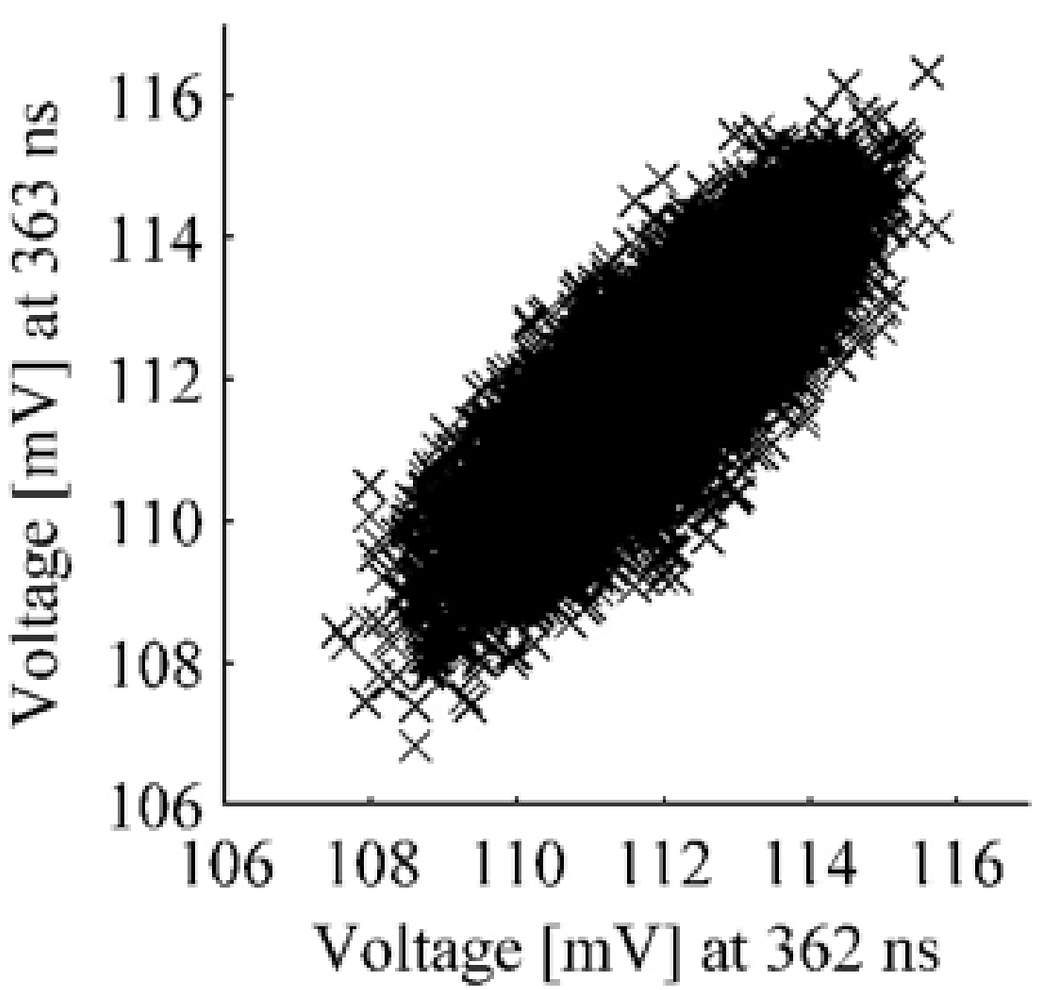


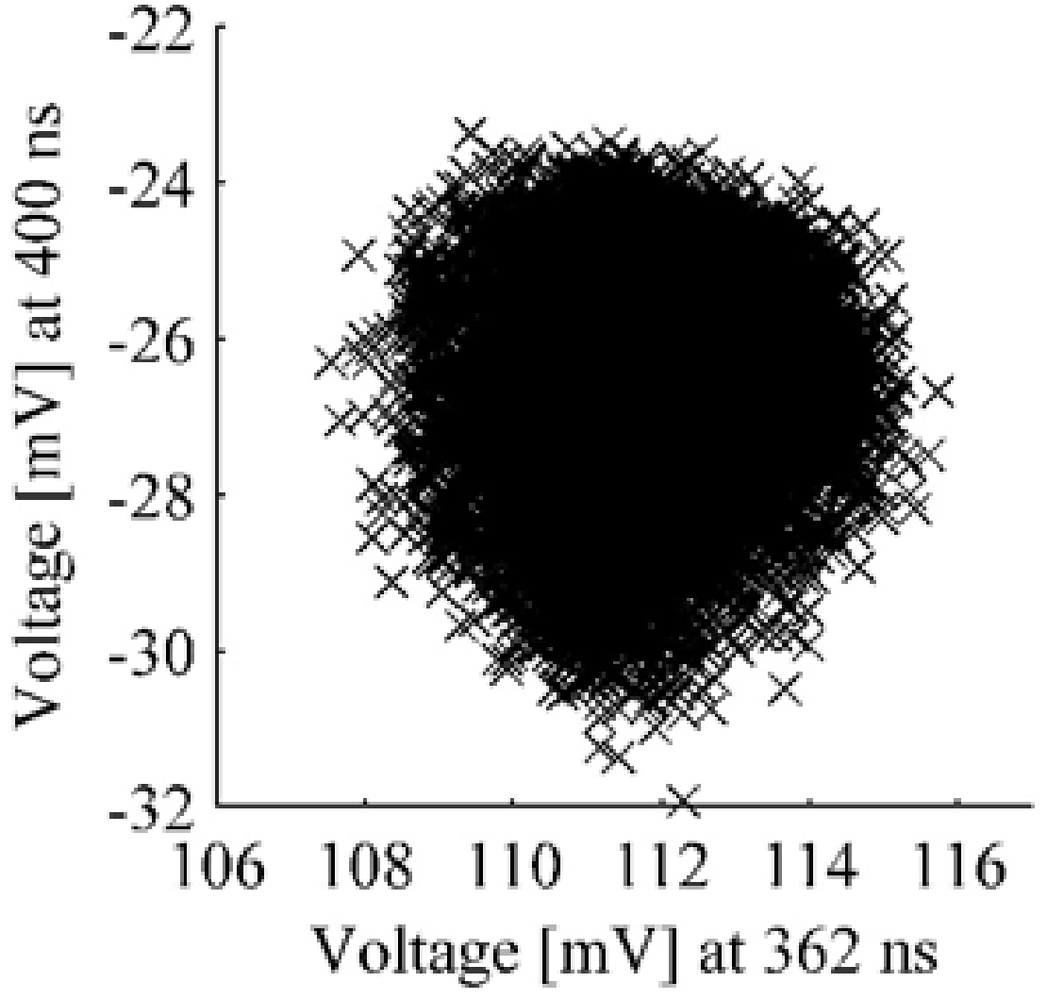




Characteristics of Power Traces

- Relationship between points





Correlation of Random Variables

- **Covariance (correlation)**

$$Cov(X,Y) = E((X - E(X))(Y - E(Y))) \\ = E(XY) - E(X)E(Y)$$

  - Estimator

$$c = \frac{1}{n-1}\sum_{i=1}^{n}(x_i - \bar{x})(y_i - \bar{y})$$

- **Correlation coefficient**

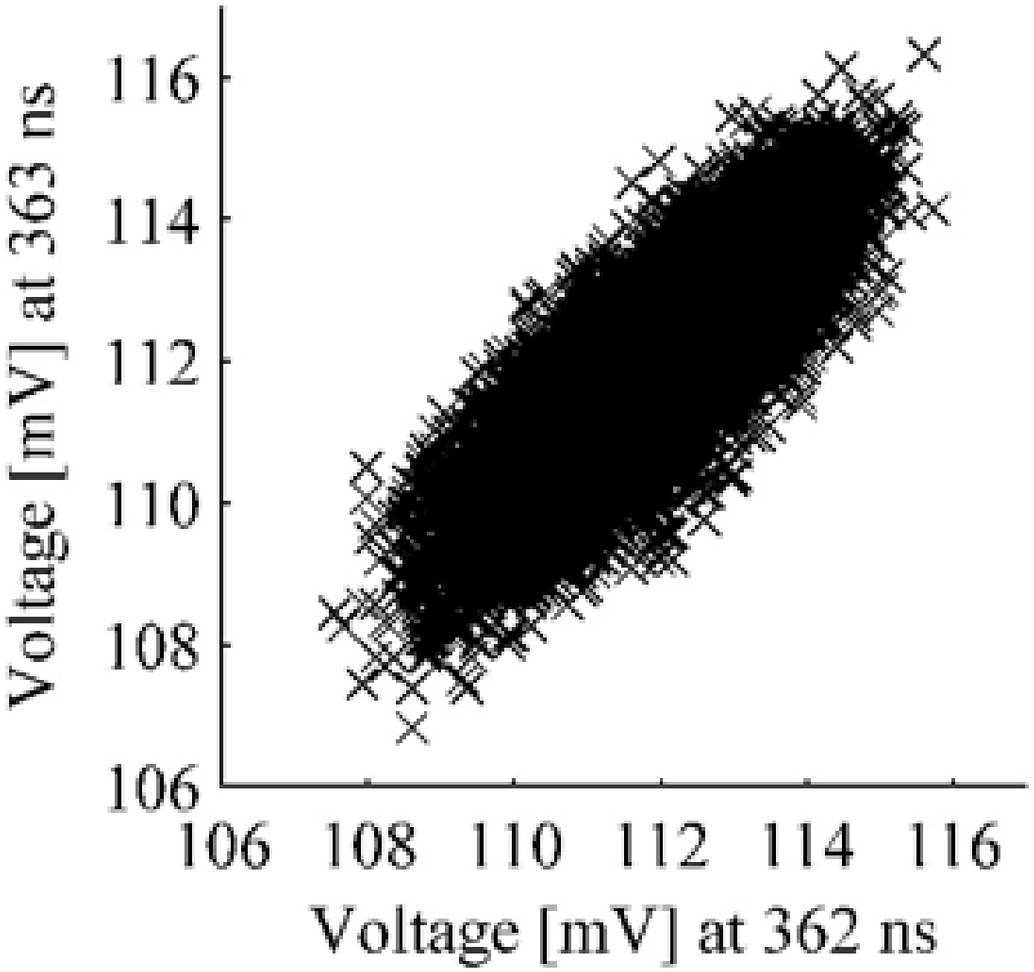
$$\rho(X,Y) = \frac{Cov(X,Y)}{\sqrt{Var(X)Var(Y)}}$$

  - Estimator

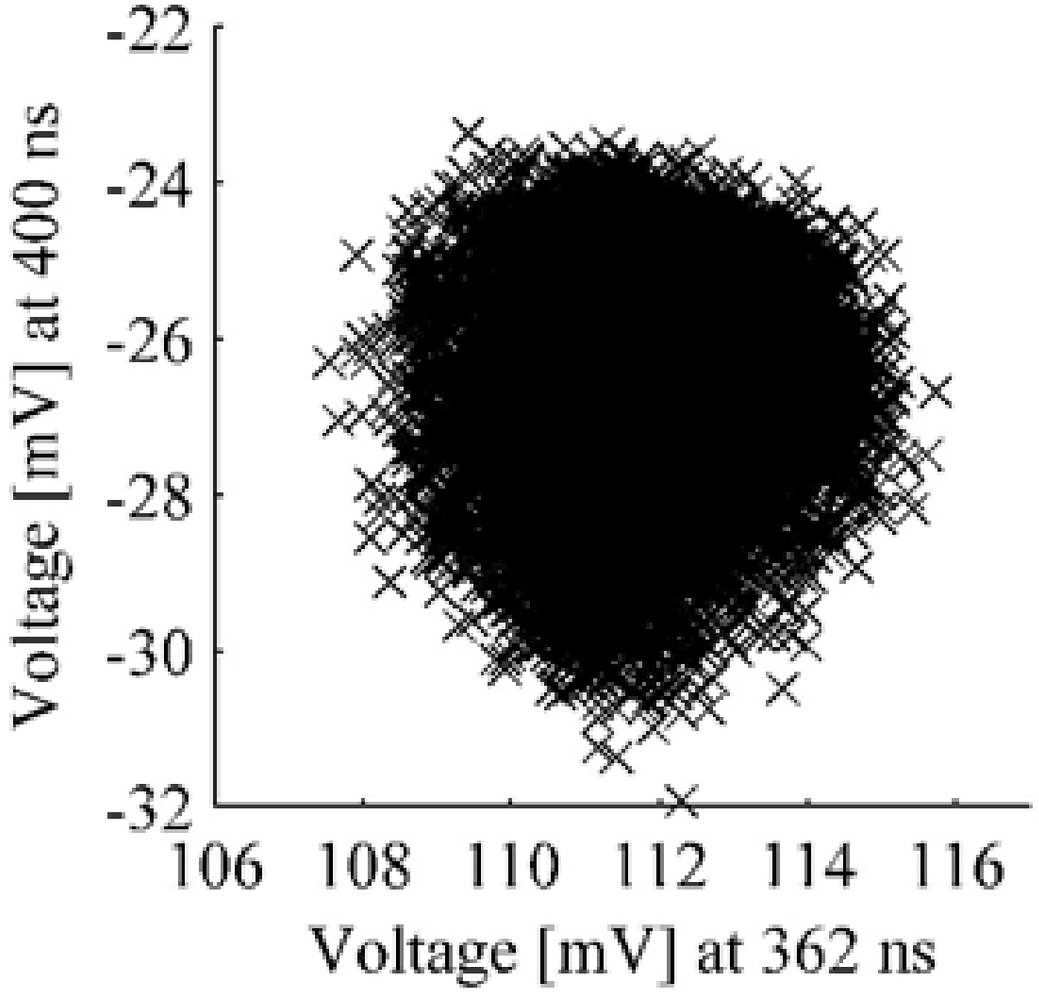
$$r = \frac{\sum_{i=1}^{n}(x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum_{i=1}^{n}(x_i - \bar{x})^2 \sum_{i=1}^{n}(y_i - \bar{y})^2}}$$

Correlation between points

$r = 0.82$



$r = 0.12$



Multivariate Gaussian Model

$$f(\mathbf{x}) = \frac{1}{\sqrt{(2\pi)^n det(\mathbf{C})}} exp\left(-\frac{1}{2}(\mathbf{x}-\mathbf{m})' \cdot \mathbf{C}^{-1} \cdot (\mathbf{x}-\mathbf{m})\right)$$

$$\mathbf{x} = (x_1, x_2, \ldots)'$$

$$\mathbf{m} = (m_1, m_2, \ldots)'$$

$$\mathbf{C} = \begin{pmatrix} c_{11} & c_{12} & \cdots \\ c_{21} & c_{22} & \cdots \\ \vdots & \vdots & \ddots \end{pmatrix}$$

covariance matrix

Compression of Traces

- **Remove the redundancy in power traces without significant loss of information**
  - Determine the points that have the highest information leakage using correlations
  - Apply compression techniques to extract useful data

Relevant Points of Power Traces

- 1000 random plaintexts
- 100 power traces for each plaintext
- Simulated number of transitions

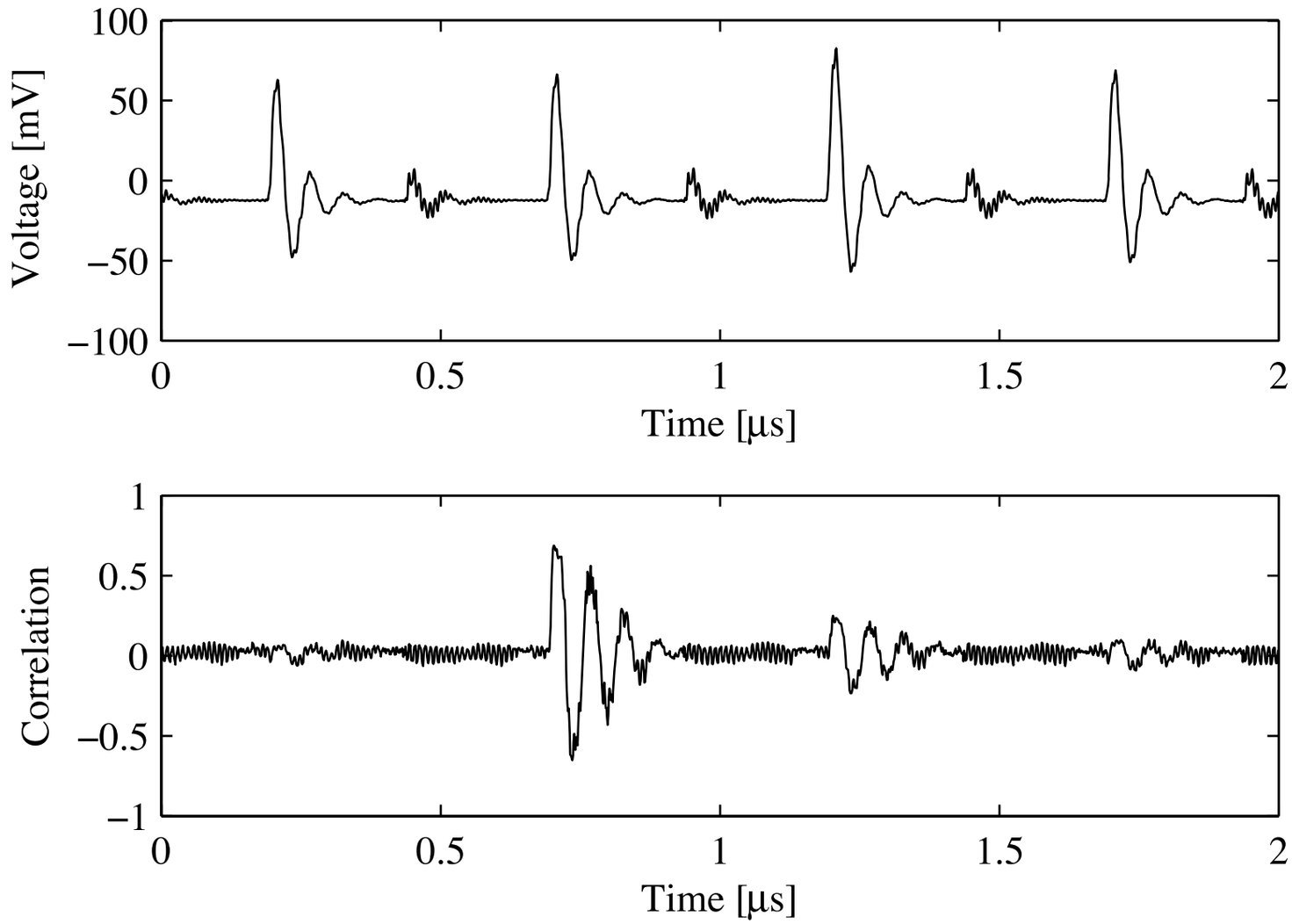


Figure 4.11. The upper plot shows a power trace of the AES ASIC during four clock cycles. The lower plot shows the correlation between the measured power traces and the simulated number of transitions that occur during the second clock cycle.

Points with Highest Information Leakage

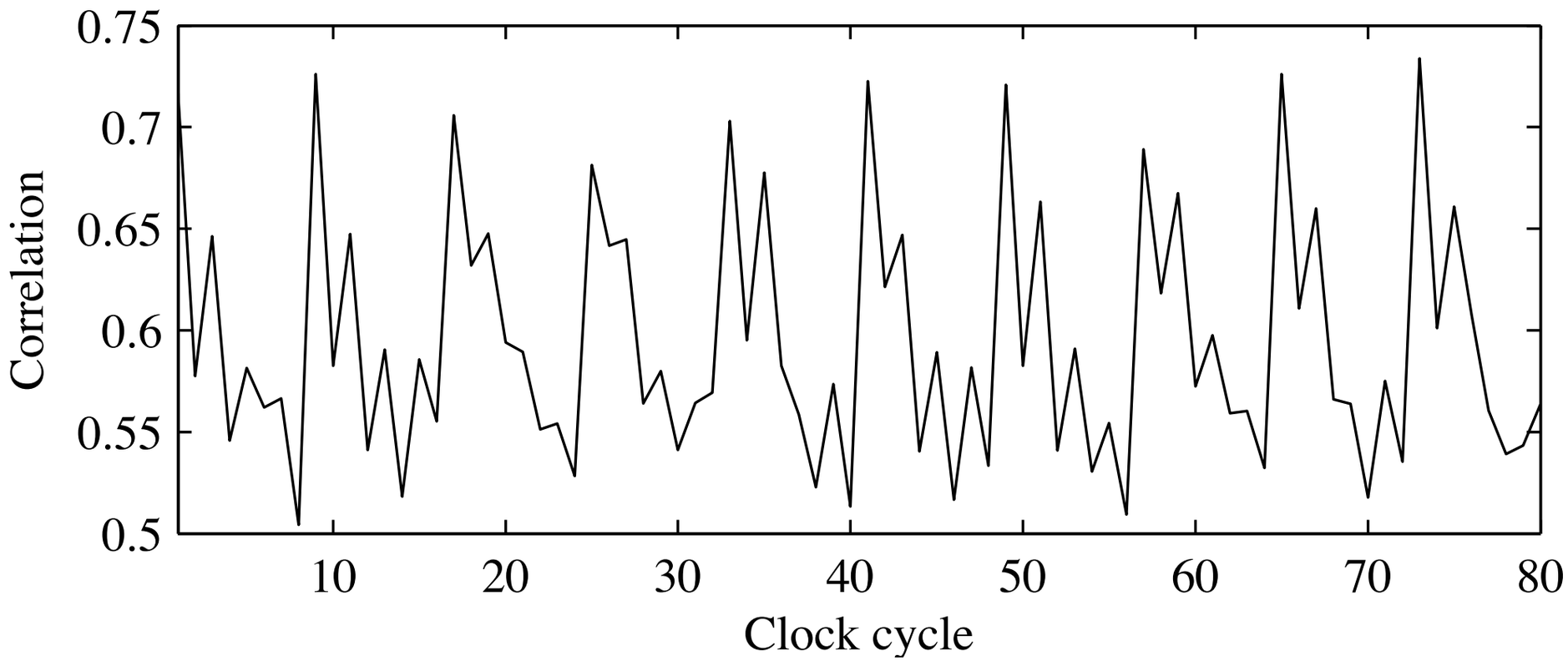


Figure 4.12. Correlation between the simulated number of transitions and the maximum peaks of the compressed power trace.

Compression Techniques

- **Maximum extraction**
- **Integration**
  - Suitable time intervals up to a clock cycle
  - Raw integration
  - Sum of absolute values
  - Sum of squares

Basic Ideas of Statistics

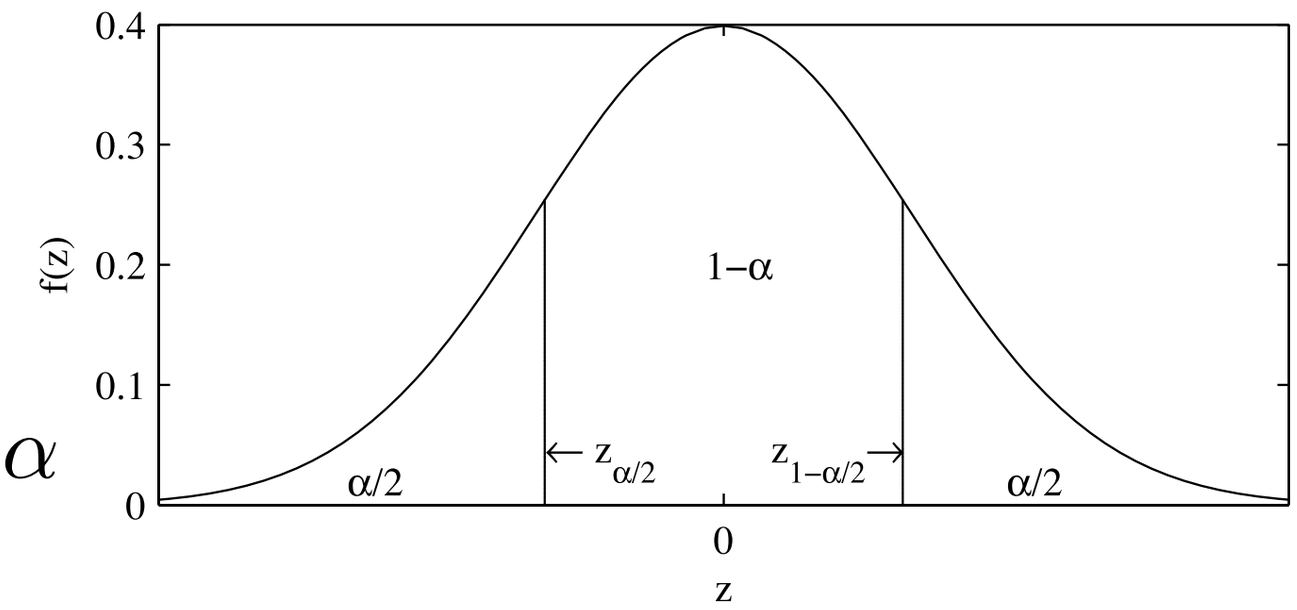
- **Sampling distribution**
- **Quality of estimations**
  - Confidence interval — quantifies how close a certain approximation is to the real parameter.
  - Hypothesis testing

Confidence Interval

- **Quantile**

$$Z \sim \mathcal{N}(0,1)$$

$$\Phi(z_\alpha) = \Pr[Z \leq z_\alpha] = \alpha$$



The interval $\left[\bar{X} - \frac{\sigma}{\sqrt{n}} z_{1-\frac{\alpha}{2}}, \bar{X} + \frac{\sigma}{\sqrt{n}} z_{1-\frac{\alpha}{2}}\right]$ is a $(1-\alpha)$ confidence interval for μ.

Hypothesis Testing

- Null hypothesis H_0
- Alternative hypothesis H_1
- Critical region
  - sets of estimated values that rejects H_0
  - complementary to the confidence interval
- **Objective: minimize the probability of "type I error", i.e. rejecting H_0 when it is true.**

Test of the Mean Value

- Hypotheses

$$H_0 : \mu = \mu_0$$
$$H_1 : \mu \neq \mu_0$$

- Probability of type I error

$$\Pr\left[|\bar{X} - \mu_0| > c\right] = \Pr\left[\frac{|\bar{X} - \mu_0|}{\sigma/\sqrt{n}} > \frac{c\sqrt{n}}{\sigma}\right] = \alpha$$

$$\Rightarrow c = \frac{\sigma}{\sqrt{n}} z_{1-\frac{\alpha}{2}}$$

$$\Rightarrow n = \left(\frac{\sigma z_{1-\frac{\alpha}{2}}}{c}\right)^2$$

Distinguishing a Parameter from Zero

- **Hypotheses**

$$H_0 : \mu = \mu_0 < 0$$
$$H_1 : \mu > 0$$

- **Number of traces**

$$\Pr\left[\bar{X} > 0\right] = \alpha$$
$$\Rightarrow \Pr\left[\frac{(\bar{X} - \mu)\sigma}{\sqrt{n}} < -\frac{\mu\sigma}{\sqrt{n}}\right] = \Phi\left(-\frac{\mu\sigma}{\sqrt{n}}\right) = 1 - \alpha$$
$$\Rightarrow n = \left(\frac{\sigma z_{1-\alpha}}{\mu}\right)^2$$

Comparing Two Sets of Traces

- Assume that

$$X = \{X_1, X_2, \ldots, X_n\}, X_i \sim \mathcal{N}(\mu_X, \sigma)$$
$$Y = \{Y_1, Y_2, \ldots, Y_m\}, Y_j \sim \mathcal{N}(\mu_Y, \sigma)$$
$$\Rightarrow \bar{X} - \bar{Y} \sim \mathcal{N}\left(\mu_X - \mu_Y, \sigma\sqrt{\frac{m+n}{mn}}\right)$$

- Confidence interval

$$\left[\bar{X} - \bar{Y} - \sigma\sqrt{\frac{m+n}{mn}}z_{1-\frac{\alpha}{2}}, \bar{X} - \bar{Y} + \sigma\sqrt{\frac{m+n}{mn}}z_{1-\frac{\alpha}{2}}\right]$$

Comparing Two Sets of Traces

- **Estimator for standard deviation**

$$s_p^2 = \frac{(n-1)s_X^2 + (m-1)s_Y^2}{m+n-2}$$

$$s_{\bar{X}-\bar{Y}} = s_p\sqrt{\frac{m+n}{mn}}$$

- **Number of traces when each set has $\frac{n}{2}$ data**

$$n = 4\left(\frac{\sigma z_{1-\frac{\alpha}{2}}}{c}\right)^2$$

Correlation Coefficient

- Fisher's transformation

$$Z_1 = \frac{1}{2} \ln \frac{1+R}{1-R}$$

$$\mu = \frac{1}{2} \ln \frac{1+\rho}{1-\rho} + \frac{\rho}{2(n-1)}$$

$$\sigma^2 = \frac{1}{n-3}$$

$$\Rightarrow Z = \frac{Z_1 - \mu}{\sigma} \sim \mathcal{N}(0,1)$$